

Cauchy-Riemann equation.

Let $f: G \rightarrow \mathbb{C}$ be a map on region G and let $U(x,y) = \operatorname{Re} f(x,y)$, $V(x,y) = \operatorname{Im} f(x,y)$. Suppose that U and V have continuous partial derivatives. Then,

f is analytic if and only if U and V satisfy the Cauchy-Riemann equation.

Proof Suppose that f is analytic. For $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+h) - f(z)}{h} = \frac{U(x+h,y) - U(x,y)}{h} + i \cdot \frac{V(x+h,y) - V(x,y)}{h}$$

Letting $h \rightarrow 0$, then $f'(z) = \frac{\partial U}{\partial x}(x,y) + i \cdot \frac{\partial V}{\partial x}(x,y)$.

Meanwhile, for $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+ih) - f(z)}{i \cdot h} = -i \cdot \frac{U(x,y+h) - U(x,y)}{h} + i \cdot \frac{V(x,y+h) - V(x,y)}{h}$$

Letting $h \rightarrow 0$, then $f'(z) = -i \cdot \frac{\partial U}{\partial y}(x,y) + \frac{\partial V}{\partial y}(x,y)$

So, $U_x = V_y$ and $V_x = -U_y$. ~~not using continuously differentiable.~~ ^{... (*)}

Conversely, U and V satisfy the condition (*), and u_x, u_y, v_x, v_y are continuous. Let $z = x + iy \in G$ be given. Since G is open, for some $r > 0$, $B_r(z) \subseteq G$. For $h = s + it \in B_r(0)$,

$$\begin{aligned} U(x+s, y+t) - U(x, y) &= [U(x+s, y+t) - U(x, y+t)] + [U(x, y+t) - U(x, y)] \\ \text{By the Mean-Value Theorem for variables } s \text{ and } t, \text{ respectively,} & \\ U(x+s, y+t) - U(x, y+t) &= U_x(x+s_1, y+t) \cdot s \text{ for some } |s_1| < |s| \\ \text{and} & \\ U(x, y+t) - U(x, y) &= U_y(x, y+t_1) \cdot t \text{ for some } |t_1| < |t| \end{aligned} \quad (*)$$

Let $Q(s, t) = [U(x+s, y+t) - U(x, y)] - [U_x(x, y) \cdot s + U_y(x, y) \cdot t]$. Now, (*) gives

$$\frac{Q(s, t)}{s + it} = \frac{s}{s + it} \cdot [U_x(x+s_1, y+t) - U_x(x, y)] + \frac{t}{s + it} \cdot [U_y(x, y+t_1) - U_y(x, y)]$$

But, $|s_1| < |s| \leq |s + it|$ and $|t_1| < |t| \leq |s + it|$, so letting $s + it \rightarrow 0$, then

$$\frac{Q(s, t)}{s + it} \rightarrow 0$$

Therefore, $U(x+s, y+t) - U(x, y) = U_x(x, y)s + U_y(x, y)t + Q(s, t)$

and similarly $V(x+s, y+t) - V(x, y) = V_x(x, y)s + V_y(x, y)t + R(s, t)$ where $\lim_{s+it \rightarrow 0} \frac{R(s, t)}{s+it} = 0$.

Now, $\frac{f(z+s+it) - f(z)}{s+it} = U_x(z) + iV_x(z) + \frac{Q(s, t) + iR(s, t)}{s+it} \rightarrow U_x(z) + iV_x(z)$.

as $s + it \rightarrow 0$.

Lemma. Let $f: G \rightarrow \mathbb{C}$ be a holomorphic at $z_0 \in G$ where G is an open set.
Then, for the differential operators

$$\frac{\partial}{\partial \bar{z}} \stackrel{\text{def}}{=} \frac{1}{2} \cdot \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \text{and} \quad \frac{\partial}{\partial z} = \frac{1}{2} \cdot \left(\frac{\partial}{\partial x} - \frac{1}{i} \cdot \frac{\partial}{\partial y} \right),$$

$$\frac{\partial f}{\partial \bar{z}}(z_0) = 0 \quad \text{and} \quad f'(z_0) = \frac{\partial f}{\partial z}(z_0) = 2 \cdot \frac{\partial u}{\partial z}(z_0),$$

Proof. Denote $f(x+iy) = u(x+iy) + i \cdot v(x+iy)$, then

$$\begin{aligned} \frac{\partial f}{\partial \bar{z}} &= \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} - \frac{1}{i} \cdot \frac{\partial u}{\partial y} \right) + \frac{1}{2} \cdot \left(i \cdot \frac{\partial v}{\partial x} - \frac{\partial v}{\partial y} \right) \\ &= \frac{1}{2} \cdot \left[\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} + i \cdot \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \right] \end{aligned}$$

so, f satisfies the Cauchy-Riemann condition iff $\frac{\partial f}{\partial \bar{z}} = 0$.

$$\begin{aligned} \text{And, } \frac{\partial f}{\partial z} &= \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} + \frac{1}{i} \cdot \frac{\partial u}{\partial y} \right) + \frac{1}{2} \cdot \left(i \cdot \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} \right) \\ &= \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \frac{i}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{aligned}$$

Polar - Coordinate form.

Thm. Let $f = u + i \cdot v$ be a Holomorphic function on $\{r \cdot e^{i\theta} \mid r > 0, -\pi < \theta \leq \pi\}$,

Then, $u_r = \frac{1}{r} \cdot v_\theta$ and $u_\theta = -\frac{1}{r} \cdot v_r$.

Proof. By the Chain Rule with respect to $z = r e^{i\theta}$, (and $\partial f / \partial \bar{z} = 0$, by Holomorphic)

$$f_r = \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial r} = \underbrace{f'(r \cdot e^{i\theta})}_{\rightarrow} \underbrace{e^{i\theta}}_{\rightarrow} \quad \text{and} \quad f_\theta = \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial \theta} = f'(r \cdot e^{i\theta}) \cdot i r e^{i\theta}$$

Therefore, $f_r = \frac{1}{i r} f_\theta$, so $u_r + i \cdot v_r = \frac{1}{i r} \cdot u_\theta + \frac{1}{r} v_\theta = \frac{1}{r} v_\theta - i \cdot r \cdot u_\theta$.

Thus, $u_r = \frac{1}{r} v_\theta$ and $v_r = -r \cdot u_\theta$, $\square$

Using this, investigate $\text{Log } z \stackrel{\text{def}}{=} \ln |z| + i \cdot \text{Arg } z$.

Given non-zero Complex number $z \in \mathbb{C}$, $z = r e^{i\theta}$ for some $r > 0$, $-\pi < \theta \leq \pi$. So,

$$\text{Log } z = \ln r + i \cdot \theta.$$

Now, the partial derivatives becomes

$$\underbrace{\frac{\partial}{\partial r} \ln r = \frac{1}{r}}_{u_r} \quad \text{and} \quad \underbrace{\frac{\partial}{\partial r} \theta = 0}_{v_r} \quad \text{and} \quad \underbrace{\frac{\partial}{\partial \theta} \ln r = 0}_{u_\theta} \quad \text{and} \quad \underbrace{\frac{\partial}{\partial \theta} \theta = 1}_{v_\theta}$$

Thus the $\text{Log } z$ satisfies the Cauchy - Riemann equation on

$$D = \{r e^{i\theta} \mid r > 0, -\pi < \theta < \pi\},$$

and the Partial derivatives of $\ln r$ and θ are continuous on D ,
Consequently, $\text{Log } z$ is Holomorphic on D .